SMART WATER USAGE & HYDRATION TRACKER

JAWAI

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- Introducti

1. Introduction

Flutter is ar
from a sing
approach fo

2. Project Overview and Requirements

The Smart
environmer
account, m
consumptio

3. Project Architecture and Structure

The Flutter
communica

```
smart_water_usage_app/
??? lib/
?   ??? main.dart             # Main entry point, UI navigation, Tracker, History & Account tabs
?   ??? analytics_screen.dart # 7-Day interactive bar charts, stats breakdown, and trends
?   ??? db_helper.dart        # Database abstraction layer (SQLite schema, CRUD, guest transfer)
??? test/
?   ??? widget_test.dart      # Smoke and UI widget tests
??? web/                      # Web platform scaffolding (includes sqlite3.wasm & sqflite_sw.js)
??? android/                  # Native Android configuration
??? linux/                    # Native Linux desktop runner
??? pubspec.yaml              # App configuration & package dependencies
??? README.md                 # Project documentation
```

Directory	P

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4. Flutter Development Environment Setup

- Downloa

- Use the integrated terminal for Flutter commands and debugging.

```
flutter --v
dart --vers
flutter doc
flutter dev
```

---

5. Dart Programming Fundamentals Used in the Project

Dart provid
classes, lis
safety.

```
Future<List<Activity>> fetchActivities() async {
  final db = await DatabaseHelper.instance.database;
  final response = await db.query('activities');

  // Convert the response into Activity objects.
  return parseActivities(response);
}
```

6. Creating and Configuring the Flutter Project

```
flutter cre
cd smart_wa
code .
flutter run
```

---

7. Smart Water Usage App ? Case Study

The Smart
is designed

- A new user opens the application and starts tracking water intake in Guest Mode.

- The user

---

8. Guest Mode and Account Management

The applica
to create a
request is s

- Display associated activity categories.

- Display da

---

9. Water Intake Tracking Module

The Water
represents

---

10. Daily Hydration Goal Module

Learning go
This distinc
hydration g

---

11. Analytics and Usage Trends Module

The Analyti

---

12. Water Log History Module

The History
and deletio

---

13. Exploring Flutter Widgets in the Project

Flutter uses
& Hydration

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14. Responsive UI Design

Responsive

```
LayoutBuilder(
  builder: (context, constraints) {
    final isWide = constraints.maxWidth > 600;

    return isWide
        ? Row(
            children: [
              Expanded(child: ProfileSection()),
              Expanded(child: SkillsSection()),
            ],
          )
        : Column(
            children: [
              ProfileSection(),
              SkillsSection(),
            ],
          );
  },
);
```

**Home Sc

---

15. Navigation Between Screens

Navigation
authenticat
to their curr

---

16. Stateful and Stateless Widgets

A `Stateles
In this proje
cards can b

- History screens update after accepting or rejecting a request.

```
setState(()
  _isLoadin
});

try {
  // ...
} finally {
  if (mount
    setStat
      _isLo
    });
  }
}
```

---

17. State Management

State mana

---

18. Custom Widgets and Styling

Reusable w

```
class CustomButton extends StatelessWidget {
  final String text;
  final VoidCallback onPressed;

  const CustomButton({
    super.key,
    required this.text,
    required this.onPressed,
  });

  @override
  Widget build(BuildContext context) {
    return ElevatedButton(
      onPressed: onPressed,
      child: Text(text),
    );
  }
}
```

19. Forms and Validation

Forms are
Validation p

---

20. Database and Data Storage

The platfor
relationship

Table	Purpose	Important Relationship

21. Charts and Data Visualization

Charts are

22. Animations and User Experience

Animations

```
AnimatedOpacity(
  opacity: isVisible ? 1.0 : 0.0,
  duration: const Duration(milliseconds: 300),
  child: const CircularProgressRing(),
);
```

Animations
database-d

---

23. Testing and Debugging

Testing ver
together.

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24. Application Workflow

```
Start / Gue
       ?
   Analytic
       ?
   History
       ?
   Account
```

- Display appropriate loading and error states.

- Avoid dup

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25. Security and Data Validation

Because th
both the Flu

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26. Conclusion

The Smart
practical m
developme
history, sta

Unit, widget, integration, and manual testing.

This docum
sample-app
documenta
goals, anal

End of Document